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Vlasov simulations of electric propulsion beam

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Abstract

A grid-based Vlasov simulation model is developed to simulate the two-dimensional unmagnetized electric propulsion (EP) plasma beam emission process. Comparing to the standard fully kinetic Particle-in-Cell simulation, the grid-based Vlasov simulation method eliminates the interference of particle noise and is capable of resolving higher-order velocity moment, such as electron heat flux, accurately. Vlasov simulations are carried out to investigate the effects of microscopic electron kinetics on macroscopic electron thermodynamics in EP beam. We find that the electron velocity distribution function (eVDF) exhibits a near-Maxwellian shape but with a depleted negative velocity tail in the beam direction and a ‘top-hat’ shape in the transverse direction. Macroscopically, the electrons confined within the quasi-neutral beam core region has a near constant temperature along the beam direction but follow a near-adiabatic cooling process as they expand outward in the transverse direction. The electron heat flux is dominated by the x -direction tensor component. The connection between the eVDF skewness and the electron heat flux suggests a pathway to develop a microscopic physics based electron closure relation for macroscopic electron thermodynamic process in unmagnetized plasma beam expansion.

Keywords: Vlasov simulation, electric propulsion beam, electron thermodynamic properties, plasma kinetics

1. Introduction

The plasma plume emitted by an electric propulsion (EP) thruster has been studied extensively. Most earlier modeling studies were concerned with plume interactions on the ion plasma time scale and focused on the dynamics of the charge-exchange (CEX) ions generated in the plume. The

majority of such studies are based on hybrid particle-in-cell (PIC) simulations where the CEX ions are modeled as macro-particles, the electrons are modeled as a massless fluid with an assumed thermodynamic closure relation, and the beam ions are modeled by an empirical density profile (see, for example, [1–6] and references therein). Recent modeling studies also addressed the dynamics of the propellant beam ions and the electrons. Such studies apply fully kinetic PIC simulations where both the beam ions and the neutralizing electrons are treated as macro-particles (see, for example [7–12] and references therein).

A major unresolved issue in previous EP plume modeling is the thermodynamic properties of the electrons and their effects on the overall plume characteristics [12]. Currently, no theoretical framework exists to describe the thermodynamic

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process in an unmagnetized collisionless plasma. Hence, a common approach in most EP plume models is to assume the electrons to be an ideal gas satisfying either the isothermal or the more general polytropic closure relation. Such an assumption leads to at least two issues. First, the electron temperature (for the isothermal assumption) or the polytropic coefficient (for the polytropic assumption), and the plume potential with respect to ambient must be provided as an input. The choice of these values can have a significant influence on the plume modeling results. Indeed, the excellent agreement between Hybrid PIC simulation [2] and in-flight measurement [13] of the Deep Space 1 ion thruster generated CEX ion environment is enabled by the availability of in-flight data of the electron temperature and plume potential. Second, and more importantly, the assumption of an equilibrium gas model for the electrons in a collisionless plasma in general is not valid because the long-range Coulomb interaction between charged particles is not an effective equilibrating mechanism. Recent fully kinetic PIC simulations of unmagnetized plasma beam emission [9–12] showed that the electrons in an ion thruster beam are non-equilibrium and anisotropic, and the electron velocity distribution function (eVDF) significantly deviates from the Maxwellian distribution. Thus, the adoption of an equilibrium gas model for electrons can lead to significant errors in plume modeling results [12].

The derivation of electron thermodynamic properties in EP plume is not trivial because the classical transport mechanisms [14] are not applicable in an unmagnetized, collisionless plasma [15]. If the electrons are strongly magnetized, the electron gyro-radius plays the role of an equivalent collisional mean free path. This can enable the construction of approximate fluid [16–19] and thermodynamic models for electrons [20]. However, if the electrons are unmagnetized, the transport mechanism is completely determined by long range interactions [21–24]. In this case, full PIC simulations of ion thruster beam and collisionless plasma expansion have shown that the macroscopic electron properties are sensitively influenced by the eVDF [10, 12, 25]. While fully kinetic PIC in theory can be applied to investigate electron thermodynamics, previous PIC simulations showed that the inherent statistical noise from the discrete particles makes it difficult to resolve accurately the higher order particle velocity moments, such as the heat flux, in practice [26].

This paper concerns with the primary beam emitted by an electrostatic thruster (e.g. ion thruster, Hall thruster). The focus is to resolve the electron kinetic and thermodynamic properties in the beam. To eliminate the interference from the inherent statistical noise of the PIC method, we present an alternative approach using the grid-based Vlasov simulation. The grid-based Vlasov simulation method has been applied in studies of fusion plasmas [27] and astrophysical plasmas [28, 29]. However, other than a few studies on Hall thruster discharge [30, 31], hollow cathode instabilities [32–34], one-dimensional plasma expansion [35–37], and magnetized beam [38], it has not been applied extensively in EP related applications. To our knowledge, this paper represents the first multi-dimensional Vlasov simulation of an unmagnetized EP beam.

The simulation study applies our recently developed Vlasov simulation code, *Vlasolver* [26, 35, 39–41]. To simplify the problem, the simulation considers the emission and expansion of a beam of cold ions and thermal electrons in a two-dimensional (2D) domain. The simulation setup does not include the initial ion and electron mixing region near the thruster exit, the CEX collisions between the beam ions and the un-ionized neutrals, and any effects from the magnetic field of the thruster or the ambient magnetic field. Section 2 discusses the grid-based Vlasov method and the simulation model setup. Section 3 first compares the results from Vlasov simulation and full particle PIC, and then discusses both the dynamics of beam expansion and the characteristics of eVDF and electron heat flux in the beam. Section 4 contains a summary and conclusions.

2. Formulation and simulation method

2.1. The Vlasov simulation method

An electrostatic, collisionless plasma system can be solved from the following Vlasov–Poisson system of partial differential equations (PDEs):

$$\frac{\partial f_\alpha}{\partial t} + \mathbf{v} \cdot \nabla_{\mathbf{x}} f_\alpha + \mathbf{a} \cdot \nabla_{\mathbf{v}} f_\alpha = 0, \quad \mathbf{a} = \frac{-q_\alpha \nabla \Phi}{m_\alpha} \quad (1)$$

$$\nabla \cdot (\epsilon_0 \nabla \Phi) = -e(n_i - n_e) \quad (2)$$

where subscript $\alpha = e$ or i denotes the electron or ion, respectively; $f_\alpha(\mathbf{x}, \mathbf{v}, t)$ the velocity distribution function (VDF); Φ the electric potential; $\mathbf{x}$ and $\mathbf{v}$ the position and velocity vector, respectively; q_α and m_α the species charge and mass, respectively; and n_i and n_e the ion and electron number density, respectively. The plasma macroscopic properties are obtained through the moments of the VDF

$$\langle F_\alpha \rangle = \int_{-\infty}^{\infty} F_\alpha \hat{f}_\alpha d\mathbf{v} \quad (3)$$

where $\hat{f}_\alpha = f_\alpha/n_\alpha$ is the normalized VDF, and $n_\alpha = \int_{-\infty}^{\infty} f_\alpha d\mathbf{v}$. We apply the following dimensionless variables in equations (1)–(3):

$$\begin{aligned} \tilde{\mathbf{x}} &= \frac{\mathbf{x}}{\lambda_{D0}}, \quad \tilde{t} = t\omega_{pe0}, \quad \tilde{\mathbf{v}}_\alpha = \frac{\mathbf{v}_\alpha}{\lambda_{D0}\omega_{pe0}}, \quad \tilde{m}_\alpha = \frac{m_\alpha}{m_e}, \quad \tilde{n}_\alpha = \frac{n_\alpha}{n_{e0}}, \\ \tilde{T}_\alpha &= \frac{k_b T_\alpha}{m_e \lambda_{D0}^2 \omega_{pe0}^2}, \\ \tilde{\Phi} &= \frac{e\Phi}{m_e \lambda_{D0}^2 \omega_{pe0}^2}, \quad \tilde{\mathbf{E}} = \frac{e\mathbf{E}}{m_e \lambda_{D0} \omega_{pe0}^2}, \quad \tilde{f}_\alpha = (\lambda_{D0} \omega_{pe0})^{N_v} f_\alpha / n_0 \end{aligned} \quad (4)$$

where e and m_e denote the electron charge and mass, respectively; n_{e0} the reference electron number density; $\lambda_{D0} = \sqrt{\epsilon_0 k_b T_{e0} / n_{e0} e^2}$ and $\omega_{pe0} = \sqrt{n_{e0} e^2 / \epsilon_0 m_e}$ the Debye length and electron plasma frequency calculated for n_{e0} and the reference temperature T_{e0} , respectively; k_b the Boltzmann constant; and T_α the temperature for specie α . The superscription N_v

denotes the velocity space dimension. In the rest of the paper, we will drop the ‘ $\sim$ ’ sign for these dimensionless variables.

In this work, we apply the ‘split’ method [42, 43] to update the Vlasov equation with a second order of accuracy in time. The Vlasov equation is solved in the phase space where the position $\mathbf{x}$ and velocity $\mathbf{v}$ are independent variables. For example, in a 2D physical space and 2D velocity space (2D2V) formulation, equation (1) can be written into

$$\frac{\partial f}{\partial t} + v_x \frac{\partial f}{\partial x} + v_y \frac{\partial f}{\partial y} + a_x \frac{\partial f}{\partial v_x} + a_y \frac{\partial f}{\partial v_y} = 0 \quad (5)$$

where a_x and a_y are determined by equation (2). Equation (5) can be split into a group of advection equations using the Strang splitting scheme

$$\frac{\partial f}{\partial t} + v_x \frac{\partial f}{\partial x} + v_y \frac{\partial f}{\partial y} = 0 \quad (6a)$$

$$\frac{\partial f}{\partial t} + a_x \frac{\partial f}{\partial v_x} + a_y \frac{\partial f}{\partial v_y} = 0. \quad (6b)$$

Equations (6a) and (6b) can be further split into a group of one-dimensional linear advection equations using the Lie splitting method.

Various time stepping schemes can be applied to solve equations (6a) and (6b). Here, the Semi-Lagrangian method is used. The numerical scheme used to solve Vlasov equation in discretized phase space is based on the third order positive flux conservation (PFC) method [44, 45]. Under the PFC scheme, for example, the splitting of one-dimensional linear advection equation along the x direction, $\partial f / \partial t + v \partial f / \partial x = 0$, can be written into following conservation form:

$$\begin{aligned} \int_{x_{i-1/2}}^{x_{i+1/2}} f(t^{n+1}, x) dx &= \Psi_{i-1/2}(t^n) \\ &+ \int_{x_{i-1/2}}^{x_{i+1/2}} f(t^n, x) dx - \Psi_{i+1/2}(t^n) \end{aligned} \quad (7)$$

where $\Psi_{i-1/2}$ and $\Psi_{i+1/2}$ are the numerical fluxes in the cells adjunct to mesh point x_i :

$$\Psi_{i+1/2}(t^n) = \int_{X(t^n, t^{n+1}, x_{i+1/2})}^{x_{i+1/2}} f(t^n, x) dx. \quad (8)$$

In equation (8), $x_{i+1/2}(t^{n+1})$ is the end of a characteristic line which starts at $X(t^n, t^{n+1}, x_{i+1/2})$. If the cell average of the function f inside a mesh cell located at x_i at time step t^n is $f_i^n = (\int_{x_{i-1/2}}^{x_{i+1/2}} f(t^n, x) dx) / \Delta x$, equation (7) can be further converted to

$$f_i^{n+1} = f_i^n + \frac{1}{\Delta x} \left(\Psi_{i-1/2}(t^n) - \Psi_{i+1/2}(t^n) \right). \quad (9)$$

The function at an arbitrary position x inside a cell is obtained through reconstruction using the 3rd order PFC method [44, 45]

$$\begin{aligned} f_h(x) &= f_i + \frac{\epsilon_i^+}{6\Delta x^2} [2(x - x_i)(x - x_{i-3/2}) \\ &+ (x - x_{i-1/2})(x - x_{i+1/2})](f_{i+1} - f_i) \\ &+ \frac{\epsilon_i^-}{6\Delta x^2} [2(x - x_i)(x - x_{i+3/2}) \\ &+ (x - x_{i-1/2})(x - x_{i+1/2})](f_i - f_{i-1}) \end{aligned} \quad (10)$$

where ϵ^+ and ϵ^- are slope correctors (see [44, 45] for definition). The numerical flux Ψ is constructed using equation (10) (see [44, 45] for the details of constructing Ψ). The PFC method guarantees the conservation of the flux and non-negative VDFs. The process to solve the Vlasov equation applies the algorithm discussed in Cheng and Knorr [42] and Mangeney and Califano [46], and is a variant of the leapfrog algorithm. One first splits the Vlasov equation into equations (6a) and (6b) and then follow the Lie splitting to update the equations in each direction. The details on the implementations of this method in our code are described in our previous work [35, 40, 41].

2.2. Simulation model and setup

We consider the emission of an unmagnetized, collisionless plasma beam of cold beam ions and thermal electrons in a 2D planar domain. Previous studies [47, 48] have shown that the differences between a 2D planar and an axi-symmetric simulation of an unmagnetized beam are reasonably small, especially in terms of the electron kinetic and thermodynamic properties. Moreover, without an external magnetic field, there is not a Lorentz force component to affect the velocity in the direction perpendicular to the 2D simulation domain. Thus, the Vlasov model uses two velocity dimensions to save computational time.

The beam flow characteristics is mesothermal, $v_{ti0} \ll v_0 \ll v_{te0}$, where v_{ti0} , v_0 , v_{te0} denote the ion thermal velocity, beam drifting velocity, and electron thermal velocity at the emission surface, respectively. Similar to that in [10], the simulations are carried out using the proton to electron mass ratio of $m_i/m_e = 1836.15$. The temperature ratio of the emitted ions and electrons at the exit is $T_{i0}/T_{e0} = 0.01$, which corresponds to a ratio of ion thermal velocity $v_{ti0} = \sqrt{T_{i0}/m_i}$ to electron thermal velocity $v_{te0} = \sqrt{T_{e0}/m_e}$ of $v_{ti0}/v_{te0} \simeq 0.0023$. The beam velocity is $v_0/v_{te0} = 0.1$. We note that these velocity and temperature ratios are similar to that of a plasma beam emitted from a typical ion thruster. For most of the results presented in this paper, the Mach number of the beam at the exit is $M_0 = v_0/C_{s0} \simeq 4.29$, where $C_{s0} = \sqrt{T_{e0}/m_i}$ is the ion acoustic velocity. (In section 3.2, results are also presented for $M_0 = 6.43$ and 8.58 to compare the effects of M_0 on beam expansion.)

An ion thruster typically uses xenon or argon plasma as the propellant. Previously, Wang and Hu [12] carried out full PIC simulations of ion thruster beam using the xenon ion to electron mass ratio of $m_i/m_e = 241073$ and the Mach number corresponding to xenon beam ions accelerated by an 1100V ion

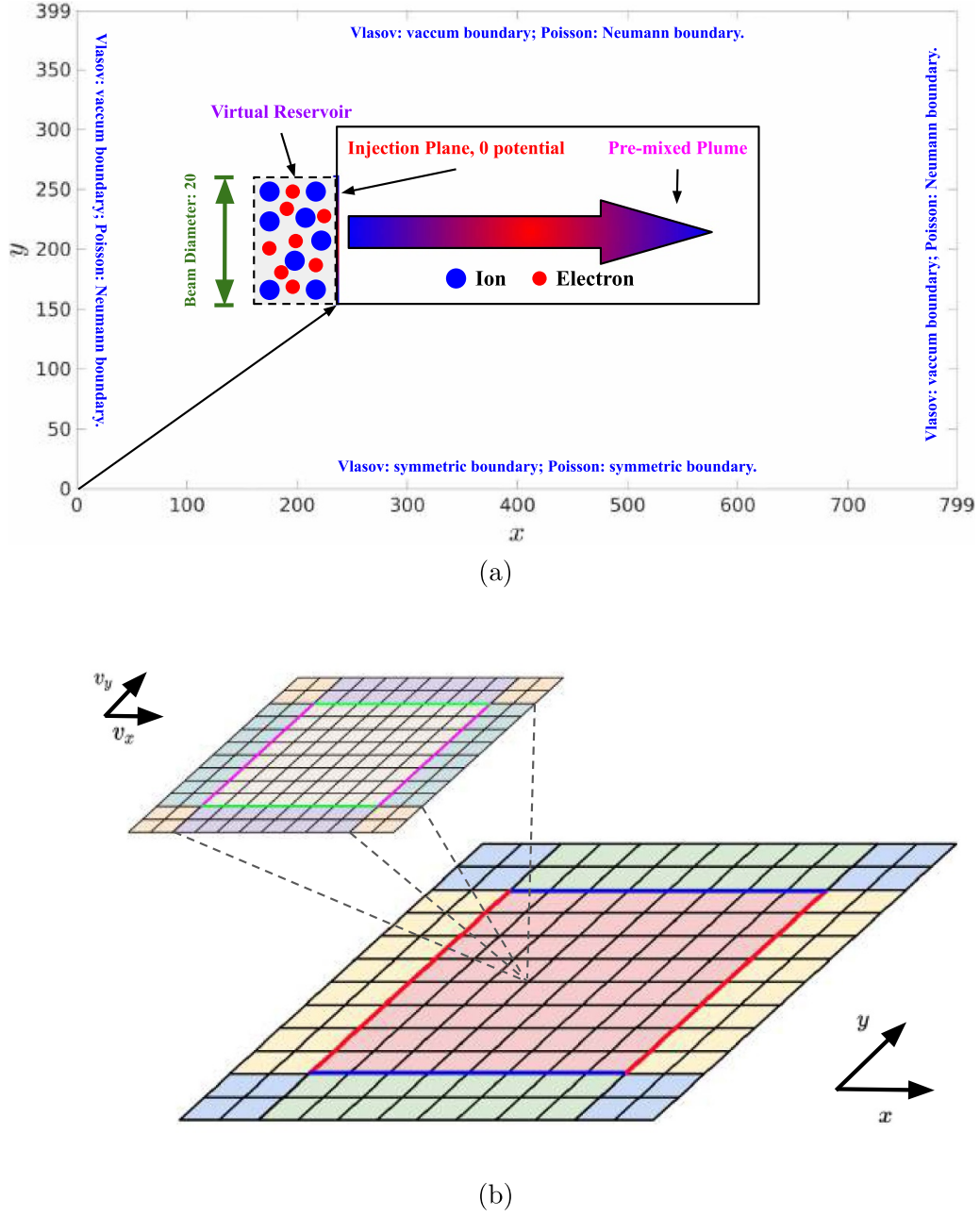


Figure 1. (a) Simulation setup. (b) Phase space configuration and discretization.

optics grid, $M_0 \simeq 20$. Hu and Wang [10, 11] also compared full PIC simulations of ion thruster beam using the ion to electron mass ratios for proton, argon, and xenon, and initial ion beam Mach numbers at $M_0 \simeq 4.29, 8.58$, and 24.55 . Those results show that the difference in ion mass do not cause a significant difference in electron properties and ion acceleration provided that the correct mesothermal velocity order is maintained in simulation. For example, Hu and Wang [11] observed less than a 3% difference in the saturated ion acceleration and electron temperature drop along the beam axis between the cases using xenon ion, argon ion, and protons. Hence, while the ion mass and M_0 used here are smaller than that in an ion thruster beam using xenon ions, the beam flow characteristics are qualitatively similar to the xenon beam [10]. The flow parameters used

are sufficient for the purpose of studying the electron properties in the beam.

Figure 1(a) shows the simulation setup. Only half of the physical domain is considered due to symmetry [40, 41]. The plasma beam is injected from an emission surface at the $x_{\min}$ domain boundary ($x=0$, $y=0-20$), with a drifting velocity v_d along the x direction. The electric potential of the emission surface is set to be $\Phi = 0$. The boundary condition for the electric field at all other boundaries is the Neumann boundary condition with $\partial\Phi/(\partial\mathbf{x} \cdot \mathbf{n}) = 0$. Figure 1(b) shows the discretized 2D2V phase space. Both the physical and the velocity space are discretized into uniform meshes. The VDF is obtained by directly solving the Vlasov-Poisson system numerically on the discretized mesh points.

The physical domain has $N_x \times N_y = 800 \times 400$ cells, with a mesh resolution of $\Delta x = \Delta y = \lambda_{D0}$. The velocity domain for the ions has $N_{i,vx} \times N_{i,vy} = 256 \times 256$ cells, with a mesh resolution $\Delta v_{i,x} = \Delta v_{i,y} = 1/640 v_{te0}$. The $v_{i,x}$ range is $-0.1 v_{te0}$ to $0.3 v_{te0}$, and the $v_{i,y}$ range is $-0.2 v_{te0}$ to $0.2 v_{te0}$. The velocity domain for the electrons has $N_{e,vx} \times N_{e,vy} = 256 \times 256$ cells, with a mesh resolution $\Delta v_{e,x} = \Delta v_{e,y} = 1/16 v_{te0}$. The $v_{e,x}$ range is $-7.9 v_{te0}$ to $8.1 v_{te0}$, and the $v_{e,y}$ range is $-8 v_{te0}$ to $8 v_{te0}$. The simulations are performed until $t\omega_{pe0} = 2000$ with a time step of $\Delta t = 0.05$ (i.e. 40 000 simulation steps). The Δt is chosen in order to satisfy the CFL condition

$$\frac{v_{\max} \Delta t}{\Delta x} \leq 1, \quad \frac{a_{\max} \Delta t}{\Delta v} \leq 1 \quad (11)$$

where $v_{\max}$ and $a_{\max}$ are the maximum velocity and acceleration, respectively. Test runs are carried out to ensure that the sizes of both the physical and velocity simulation domain are sufficiently large so that the beam expansion is still well within the simulation domain at the end of simulation and there is no interference from the domain boundary on the simulation results. While the simulation is carried out as a transient simulation, the simulation time is long enough so that the beam properties have reached a steady state behind the propagating beam front.

An EP device emits equal ion and electron fluxes which form a quasi-neutral beam outside the thruster exit. In the simulation, we enforce the quasi-neutral ($n_i = n_e$) and current free ($J_i = J_e$) conditions at the emission exit. The quasi-neutral and current free emitting condition is achieved through the use of a virtual reservoir placed within the ghost cells adjacent to the emission plane. Inside the reservoir, the electrons and ions are pre-loaded with the Maxwellian VDF. The ‘particles’ with $v_x \geq 0$ are injected into the simulation domain. While the cold ions rarely back flow from the emission surface, a large population of the electrons may back flow due to their much larger mobility. Hence, the number density of the injected thermal electrons is dynamically adjusted based on the electron to ion density ratio just downstream of the emission surface,

$$n_e^* = (n_{i,\text{exit}}/n_{e,\text{exit}}) n_0. \quad (12)$$

Thus, the VDF of the ions and electrons at the emission plane are

$$f_i(v_x, v_y) = n_0 \frac{1}{(\sqrt{2\pi} v_{ti0})^2} \exp\left(-\frac{(v - v_d)^2}{2v_{ti0}^2}\right), \quad v_x \geq 0 \quad (13a)$$

$$f_e(v_x, v_y) = n_e^* \frac{1}{(\sqrt{2\pi} v_{te0})^2} \exp\left(-\frac{(v - v_d)^2}{2v_{te0}^2}\right), \quad v_x \geq 0. \quad (13b)$$

At all other physical boundaries except the $y = 0$ boundary, we apply the vacuum boundary condition implemented by setting zero to the VDF in all of the ghost cells adjacent to the boundaries.

$$\begin{aligned} f_{[(x=N_x+1, y); (v_x, v_y)]} &= 0, f_{[(x=N_x+2, y); (v_x, v_y)]} = 0, \\ f_{[(x=N_x+3, y); (v_x, v_y)]} &= 0. \end{aligned} \quad (14)$$

At the boundary at $y = 0$, the specular reflection method from [49] is adopted for the symmetric boundary condition. The symmetric boundary condition is implemented as [49, 50]

$$\begin{aligned} f_{[(x, y=-3); (v_x, v_y=ivy)]} &= f_{[(x, y=2); (v_x, v_y=N_{iy}-ivy)]} \\ f_{[(x, y=-2); (v_x, v_y=ivy)]} &= f_{[(x, y=1); (v_x, v_y=N_{iy}-ivy)]} \\ f_{[(x, y=-1); (v_x, v_y=ivy)]} &= f_{[(x, y=0); (v_x, v_y=N_{iy}-ivy)]}. \end{aligned} \quad (15)$$

In the velocity domain, a ‘cut-off’ boundary condition is implemented as we find the selected velocity range is sufficiently large such that the VDF approaches zero at the velocity space boundary.

$$f_{[(x, y); (v_x > v_{x,\max}, v_y)]} = 0 \quad (16a)$$

$$f_{[(x, y); (v_x, v_y > v_{y,\max})]} = 0. \quad (16b)$$

The *Vlasolver* code is fully parallelized. The simulations are carried out on the USC CARC’s Discovery cluster. A typical run uses around 40 billion mesh points and requires about 800 CPU cores. A simulation running for 40 000 numerical steps typically finishes within 98 hours wall clock.

3. Results and discussions

3.1. Comparison with fully kinetic PIC simulations

Previously, Wang and Hu [12] and Hu and Wang [10, 11] carried out full particle PIC simulations of ion thruster beam emission. In this section, we first compare Vlasov simulation with previous full particle simulations using the same setup as that in [10, 11]. In [10, 11], PIC simulations were carried out using a number particles per cell (N_p) of $N_p = O(10^2)$ at the beam exit. Here, to suppress numerical noise in the low density region, we rerun the simulations presented in [10, 11] using $N_p = O(10^4)$ at the beam exit. As the statistical noise in PIC scales inversely with $\sqrt{N_p}$, the use of 10^4 particles per cell at the injection plane will reduce the noise level in the results of [10, 11] by one order of magnitude.

Figure 2 shows the comparison of ion density, electron density, and potential between the results from Vlasov and PIC simulations at $t\omega_{pe0} = 1000$. In an EP beam, the cold beam ions are confined within a well defined beam boundary while the much more mobile thermal electrons can propagate well beyond this beam boundary. We find that the beam remains quasi-neutral until the density is reduced by expansion to $n_{\text{front}} \simeq 0.016$. Beyond the location of $n_i \simeq n_e \sim n_{\text{front}}$, n_i quickly falls to near zero while n_e gradually falls to zero over a large region. (In figure 2(a), the outermost dashed black contour line is at $n = 0.016$ and the magenta dashed line is at $n = 5 \times 10^{-4}$.) The trajectory for the ion emitted at the edge of the beam exit ($x = 0, y = 20$) is also shown in figure 2(a),

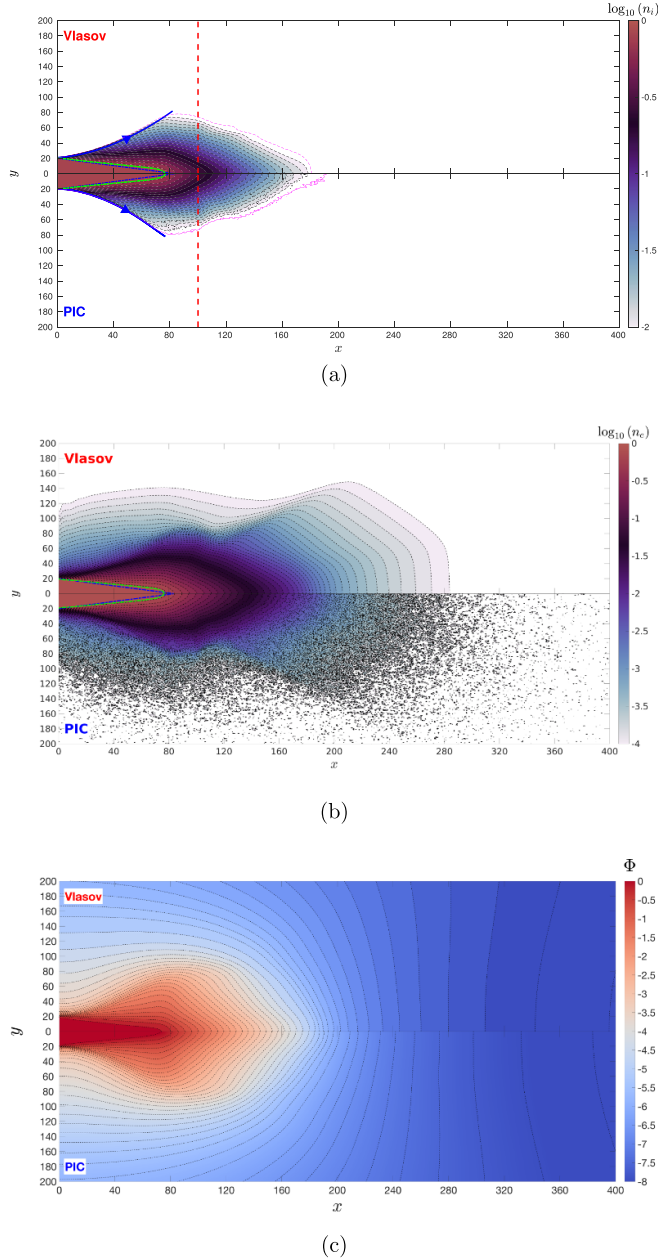


Figure 2. Comparisons of simulation results between PIC and Vlasov at $t = 1000$. (a) Ion density contour. The red dashed line shows the approximate estimated beam front location on the beam axis, $x = v_d t$. The blue arrowed line shows the trajectory of ion starting from $(x = 0, y = 20)$. The blue dash-dotted line shows the first Mach Line. The magenta dashed line shows the iso-contour line of $n_i = 5 \times 10^{-4}$. The iso-contour line of n_i on-axis rarefaction wave front is shown using green dash-dotted line. (b) Electron density contour. The blue dash-dotted line shows the first Mach Line. The iso-contour line of n_e on-axis rarefaction wave front is shown using green dash-dotted line. (c) Potential contour.

which agrees well with the $n_i \simeq n_{\text{front}}$ iso-density line on the beam side. In this paper, we refer the region where the beam is quasi-neutral as the *quasi-neutral beam region*, with the boundary defined by the iso-density line of $n_i \simeq n_e \simeq n_{\text{front}}$. We refer the region beyond the ion beam boundary as the *electron gas region*.

We find that the ion density contours from Vlasov and PIC are almost identical. This is to be expected because, in the PIC simulation, the number of particles per cell in the ion beam region is more than $O(10^3)$. Thus, there is little statistical noise effect in the PIC result. On the other hand, while the electron density contours from Vlasov and PIC agree well in the ion beam region, the electron density contours in PIC start to become noisy in the electron gas region when the number of particles per cell falls to $O(1) - O(10)$. We note that the PIC simulation cannot resolve the outer boundary of the electron gas region because of the particle noise. On the other hand, the Vlasov simulation shows very smooth n_e contours all the way to $n_e \sim 0$. The electric potential contours from Vlasov and PIC also agree well within the ion beam region but show a small discrepancy in the low density electron gas region due to the effects of particle noise on the charge density in PIC.

The blue dash dotted line shown in figures 2(a) and (b) is the Mach line drawn from the edge of the beam exit ($x = 0$, $y_{\text{edge}} = 20$):

$$y = y_{\text{edge}} - \frac{x}{\sqrt{M_0^2 - 1}}. \quad (17)$$

Previous analytical study and hybrid PIC simulation of mesothermal plasma expansion [51, 52] showed that, similar to that in compressible supersonic flow, the Mach line is the boundary that separates the unperturbed flow region from the expansion region. The simulation result agrees with the previous theoretical solution [51, 52]. The region within the Mach line has a uniform plasma density, with $n_i \simeq n_e \simeq n_0$, and will be referred to as the *unperturbed beam core region*.

Figure 3 compares the electron temperature T_e contours between Vlasov and PIC simulation results. The electron temperature in Vlasov simulation is obtained by directly integrating the electron VDF at every mesh points

$$T_{e,k} = \frac{1}{n_e} \iint m_e (v_k - \langle v_k \rangle)^2 f_e dv_x dv_y \quad (18)$$

where subscript k denotes the direction x or y . The electron temperature in PIC simulation is obtained from the macro-particles within each cell [10]. The results are calculated for mesh points with a density of $n_e \geq 5 \times 10^{-4}$. The comparison shows generally good agreement between the two methods. The PIC result becomes noisy at locations where the density is below $n_e \sim 10^{-2}$. At these locations, the particle per cell is at or lower than $O(10^2)$. On the other hand, the Vlasov simulation result shows a smooth temperature contour throughout the electron gas domain.

Figure 4 compares the electron heat flux vectors between the Vlasov and PIC simulation results. The electron heat flux vector $Q_{e,n}$ and heat flux tensor $Q_{e,kmn}$ are defined as

$$\begin{aligned} Q_{e,kmn} &= \iint m_e (v_{e,k} - \langle v_{e,k} \rangle) (v_{e,m} - \langle v_{e,m} \rangle) \\ &\quad \times (v_{e,n} - \langle v_{e,n} \rangle) f_e dv_x dv_y \\ Q_{e,n} &= \iint \frac{1}{2} m_e (v - \langle v \rangle)^2 (v_{e,n} - \langle v_{e,n} \rangle) f_e dv_x dv_y \end{aligned} \quad (19)$$

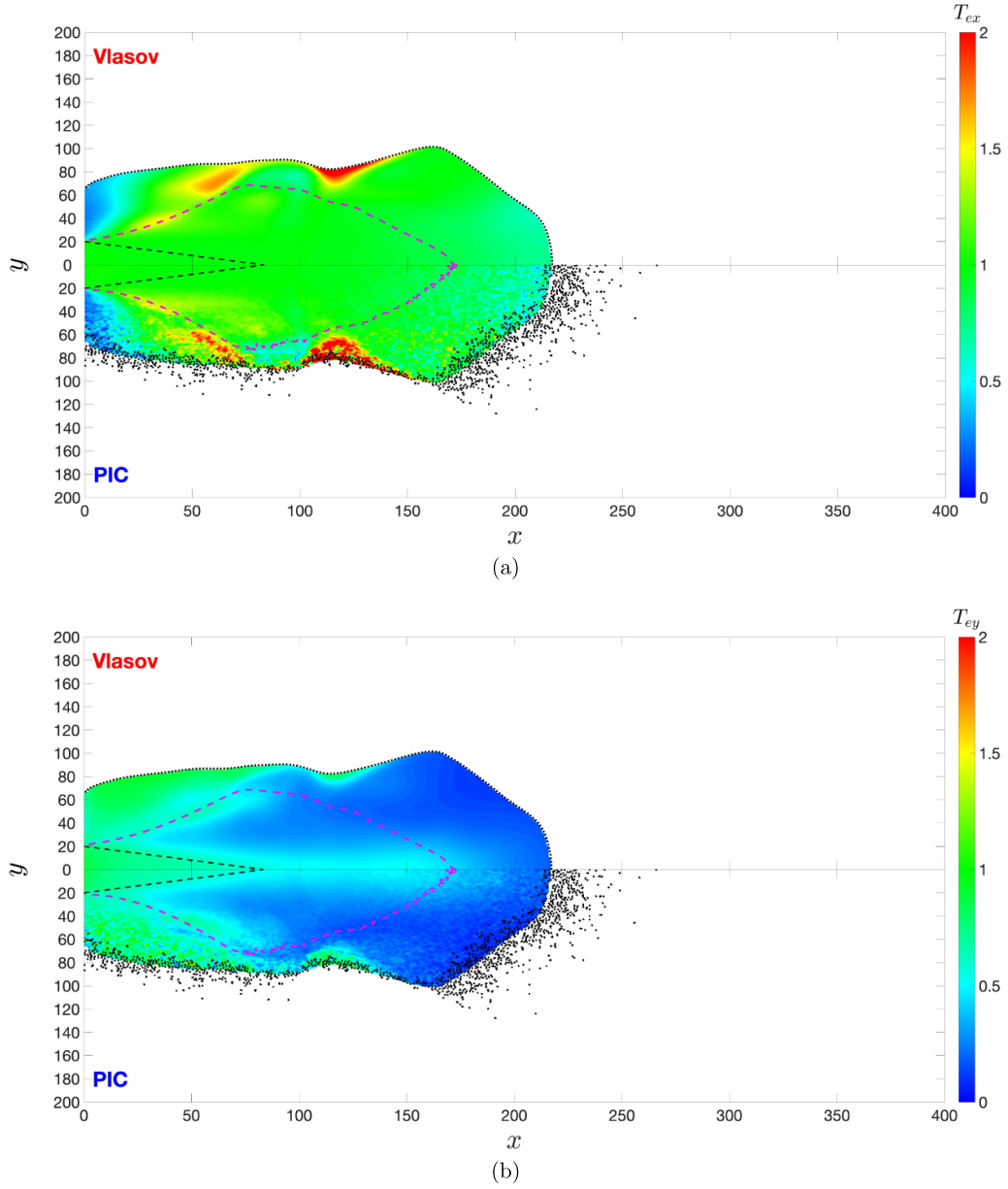


Figure 3. Comparisons of simulation results at $t = 1000$ on electron temperature on both x and y direction between grid-based Vlasov method and particle-based PIC methods. The black dashed line shows the first Mach line. The magenta dashed line shows the ion beam envelope. The black dotted line shows the iso-contour line of $n_e \sim 0$. (a) Electron x direction temperature T_{ex} . (b) Electron y direction temperature T_{ey} .

where subscripts k , m and n denote the direction x or y . The heat flux in Vlasov simulation is obtained directly from equation (19). The heat flux in PIC simulation is obtained from the macro-particles within each cell [10]. The results are calculated for mesh points with a density of $n_e \geq 5 \times 10^{-4}$. We find that the heat flux contours are qualitatively similar between the two methods. However, the heat flux obtained from PIC is noisy even in the ‘unperturbed beam core’ region, where $n_e \sim O(1)$ and $N_p \sim O(10^4)$. As heat flux is the third order velocity moment, the effects from the discrete particles noise

are amplified in the PIC result. On the other hand, the heat flux obtained from Vlasov remains smooth in the entire simulation domain.

Figures 5 and 6 compare the local eVDFs in the v_x and v_y directions, respectively. The eVDFs plotted are selected from three beam locations, where the local number of particles per cell in PIC simulation is approximately at $N_p \sim O(10^4)$, $O(10^3)$, and $O(10^2)$, respectively. The corresponding Maxwellian distributions, reconstructed from the local electron temperature and electron bulk velocity, are also included

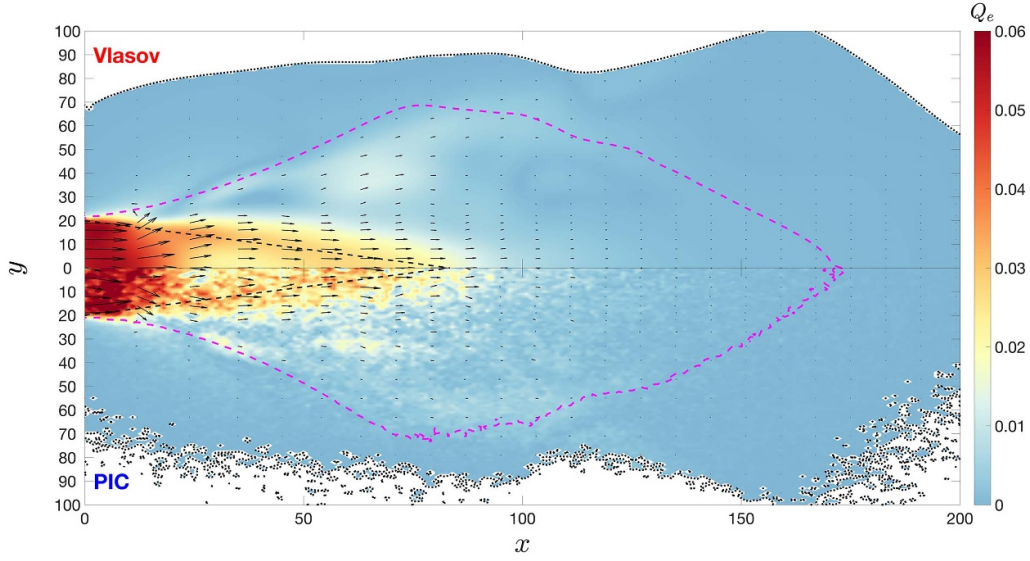


Figure 4. Comparisons of simulation results at $t = 1000$ on electron heat flux Q_{ea} between grid-based Vlasov method and particle-based PIC methods. The black dashed line shows the first Mach line. The magenta dashed line shows the ion beam envelope. The black dotted line shows the iso-contour line of $n_e \sim 0$.

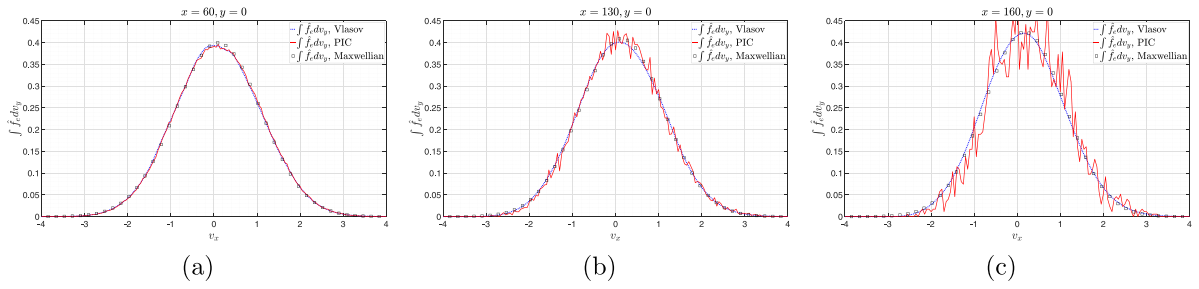


Figure 5. Comparison of simulation results at $t = 1000$ on density normalized eVDF vs v_x between Vlasov and PIC methods. The red solid lines show the eVDF plot obtained from PIC simulations. The blue dotted lines show the eVDF plot obtained from the grid-based Vlasov simulations. The black square shows the corresponding Maxwellian distribution reconstructed from the local temperature and bulk velocity. Panels (a)–(c) show eVDFs at locations where PIC has $N_p \sim O(10^4)$, $O(10^3)$, and $O(10^2)$ respectively.

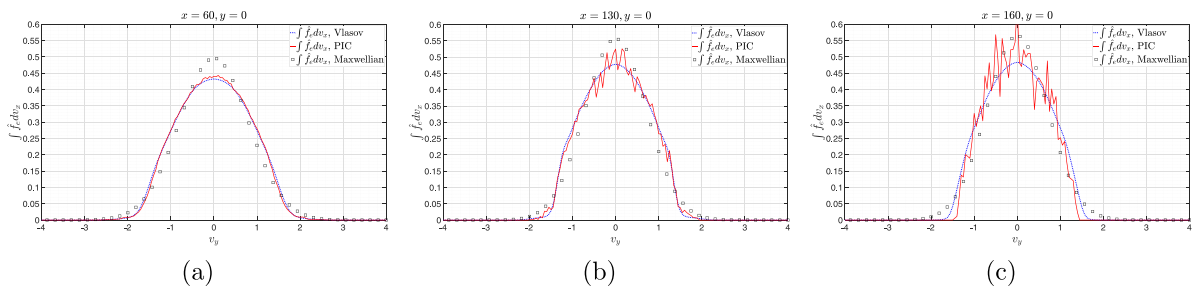


Figure 6. Comparison of simulation results at $t = 1000$ on density normalized eVDF vs v_y between Vlasov and PIC methods. The red solid lines show the eVDF plot obtained from PIC simulations. The blue dotted lines show the eVDF plot obtained from the grid-based Vlasov simulations. The black square shows the corresponding Maxwellian distribution reconstructed from the local temperature and bulk velocity. Panels (a) to (c) show eVDFs at locations where PIC has $N_p \sim O(10^4)$, $O(10^3)$, and $O(10^2)$ respectively.

in the plot for comparison. The eVDFs obtained from both PIC and Vlasov simulation exhibit similar features. At $N_p \sim O(10^4)$, the eVDFs from PIC have a low level of numerical noise and closely matches the Vlasov simulation results. However, at $N_p < O(10^3)$, the eVDFs from PIC show

substantial fluctuations around the eVDF from Vlasov simulation due to numerical noise.

The comparison of macroscopic properties shows that, in order to suppress the influence of statistical particle noise on density, temperature, and heat flux in PIC simulations,

it is necessary to maintain a local particle per cell of $N_p \geq O(10)$, $N_p \geq O(10^2)$, and $N_p \geq O(10^4)$, respectively. Given that maintaining a local $N_p \geq O(10^4)$ particles per cell is computationally prohibitive for most practical PIC studies, and that the statistical particle noise in PIC can lead to artificial relaxation and heating [53–55], it is therefore not efficient to apply PIC simulations to accurately investigate the electron thermodynamic properties in an expanding EP plume.

3.2. Quasi-neutral beam front and ion rarefaction wave front in the beam

The one-dimensional (1D) expansion of a semi-infinite collisionless plasma into vacuum is a classical problem, and has been studied extensively [56]. The 1D expansion at ion plasma time scale generates a forward propagating quasi-neutral plasma beam front into the vacuum and a backward propagating ion rarefaction wave, at ion acoustic speed, into the unperturbed plasma. For a plasma beam with an injection velocity v_0 or a Mach number $M_0 = v_0/C_{s0}$ along the x direction, the normalized location of the forward propagating quasi-neutral plasma beam front is given by [35, 57]

$$x_+^{\text{ion}}(t) - x_0 \simeq \frac{2t}{\sqrt{m_i/m_e}} \ln \left(\alpha \cdot \frac{t}{\sqrt{m_i/m_e}} \right) + \frac{M_0 \cdot t}{\sqrt{m_i/m_e}} \quad (20)$$

and that of the backward propagating ion rarefaction wave front is given by [11, 56]

$$x_-^{\text{ion}}(t) - x_0 = \frac{t}{\sqrt{m_i/m_e}} (M_0 - 1) \quad (21)$$

where α is a numerical fitting parameter, and x_0 is the plasma emission surface. Note that in equations (20) and (21), $t/\sqrt{m_i/m_e}$ is the time normalized by ion plasma frequency. Medvedev [57] found that $\alpha = 0.19$ provided the best fit for the quasi-neutral expansion front in 1D expansion. (In our simulation, we found that a slightly adjusted value of $\alpha = 0.20$ provided the best fit in 2D beam expansion).

Mora and Grismayer [58] and Grismayer *et al* [59] carried out analytical and numerical studies of 1D expansion of a plasma slab, and showed that during the initial phase of expansion ($1 \ll t/\sqrt{m_i/m_e} \ll L/\lambda_{D0}$, where L is the width of the initial plasma slab), the expansion on both sides of the slab can be described by the self-similar solution for the semi-infinite plasma expansion process [58, 59]. In this phase, two ion rarefaction waves propagate inward toward the center of the plasma slab. As the expansion progresses, the two ion rarefaction wave fronts eventually meet at the center of the slab. Beyond this point, the expansion process can no longer be accurately described by the self-similar solution for 1D semi-infinite plasma expansion. Khrabrov *et al* [60] extended this analysis to a bounded one-dimensional plasma slab, and developed an implicit analytical solution to describe further evolution after the rarefaction wave fronts meet at the center.

The overall EP beam structure is determined by the coupled expansion processes along the beam and transverse to the

beam. The expansion along the beam exhibits some of the similar characteristics of the 1D semi-infinite plasma expansion [56], and the expansion transverse to the beam exhibits some of the similar characteristics of the 1D finite-size plasma slab expansion [58, 59]. Figure 7 shows the potential profiles along the beam axis in the 2D beam emission obtained from Vlasov simulation at three different times. The Vlasov simulation results resolve both the self-similar solution of beam front expansion and the ion rarefaction wave propagating opposite to the expansion direction along the beam direction. The dotted vertical lines in figure 7 show the location of the on-axis quasi-neutral plasma expansion front $x_+^{\text{ion}}(t)$ given by equation (20) (where $\alpha = 0.20$ is used as the fitting parameter) and the dashed vertical lines show the location of the on-axis rarefaction wave front $x_-^{\text{ion}}(t)$ given by equation (21). We find that both the quasi-neutral beam front and the rarefaction wave front obtained from Vlasov simulation match that given by the analytical solutions. Both the backward propagating ion rarefaction wave front and the forward propagating quasi-neutral expansion front are contained within the quasi-neutral beam region.

To investigate the effects of the initial Mach number M_0 on the coupling between the beam direction expansion and the transverse direction expansion, we carry out simulations using different initial Mach numbers while keeping all other parameters unchanged. Figure 8 compares the electric potential contours for $M_0 = 4.29, 6.43$, and 8.58 . The iso-contour line of the electron density at the on-axis rarefaction wave front and the iso-contour line of the ion density at the beam front (ion beam envelope) are also shown, along with the iso-contour line for $n_e \sim 0$. The results show that while the structure of the transverse expansion extends further along the beam axis direction at a larger M_0 , the overall expansion features remain similar. Therefore, in the following discussions, we will focus on the base simulation case using $M_0 = 4.29$.

3.3. Electron velocity distribution function

The grid-based Vlasov method directly obtains the VDF at each grid point in the simulation domain. We next examine the eVDF in different regions in the beam: the unperturbed beam core region, the quasi-neutral beam region, and the electron gas region. The locations where the eVDFs are plotted are shown in figure 9, and the contours of the normalized eVDF, $\hat{f}_e = f_e/n_0$, are shown in figure 10.

Figure 10(a) shows the eVDF at a location within the unperturbed beam core region, $\hat{f}_e(x = 40, y = 0)$. At the emission exit, only half of the velocity space is initially filled by the drifting Maxwellian VDF, equation (13). As the electrons are reflected from the downstream region, the initially empty half of the velocity space becomes populated. The high-energy tail in the v_y direction is also reduced slightly. Figure 10(b) shows the eVDF at a location in the quasi-neutral beam region, $\hat{f}_e(x = 100, y = 0)$. In the v_x direction, the VDF exhibits a near-Maxwellian shape. However, in the v_y direction, the VDF shows a further reduced population in the high-energy tail and an increased population in the low-energy regions. Figure 10(c) shows the VDF near the on-axis

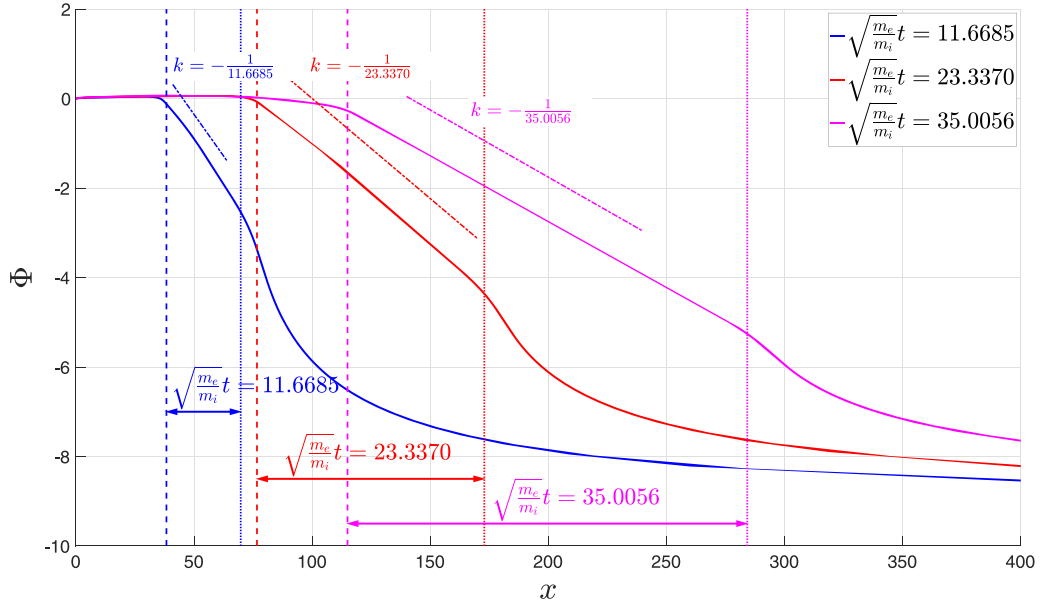


Figure 7. 1D potential profile along the beam axis $y = 0$ at different time. Vertical dashed line: on-axis ion rarefaction wave front, equation (21); Vertical dotted line: on-axis quasi-neutral expansion front, equation (20). Note that $\sqrt{m_e/m_i} \cdot t$ is the normalized time at ion plasma scale.

ion beam front, $\hat{f}_e(x = 160, y = 0)$. Comparing to $\hat{f}_e(x = 100, y = 0)$, $\hat{f}_e(x = 160, y = 0)$ has a smaller high-energy tails in the v_y direction within the ($v_x > 0, v_y$) velocity space. This is because a double layer forms near the ion beam front [57, 61], leading to an enhanced electric field along the beam direction. Consequently, only those electrons with a sufficiently large v_x can reach the ion beam front. At the same kinetic energy, those electrons will have a relatively smaller v_y . Figure 10(d) shows the eVDF beyond the ion beam front, the electron gas region, $\hat{f}_e(x = 220, y = 0)$. Wang *et al* [8] showed that a potential well forms between the ion beam front and the beam emission plane, effectively trapping the majority of emitted thermal electrons. Only those electrons with sufficiently high energy are able to overcome the potential well and escape the ion beam front.

We integrate the eVDF over v_y to obtain the 1D eVDF in the v_x direction, $\hat{f}_e(v_x) = \int_{-\infty}^{\infty} \hat{f}_e dv_y$. Figures 11(a)–(d) show $\hat{f}_e(v_x)$ along the beam axis, $y = 0$, and figures 11(e)–(h) show $\hat{f}_e(v_x)$ at the same x positions along the off-center line of $y = 20$. For comparison, a drifting Maxwellian VDF based on the local temperature and bulk velocity is also plotted. We find that, while the core region of the $\hat{f}_e(v_x)$ maintains the Maxwellian shape, a depletion of the electrons in the $-v_x$ tail develops along the beam direction. This is due to the reflection of the electrons by the ambipolar field at the ion beam front. In the region beyond the ion beam front, a slight increase is also observed in the $+v_x$ tail because only the high-energy electrons are able to escape the ion beam front.

We next integrate the eVDF over v_x to obtain the 1D eVDF in the v_y direction, $\hat{f}_e(v_y) = \int_{-\infty}^{\infty} \hat{f}_e dv_x$. Figure 12 shows $\hat{f}_e(v_y)$ at various locations along the beam axis, $y = 0$. For comparison, a Maxwellian VDF based on the local temperature

$T_{e,y}(x, y)$ is also plotted. We find that $\hat{f}_e(v_y)$ significantly deviates from the Maxwellian VDF.

To understand the formation of the non-Maxwellian eVDF in 2D beam expansion, we conduct a comparative analysis between the spatial evolution of $\hat{f}_e(v_y)$ and the temporal evolution of eVDF in a 1D finite-size plasma expansion. Mora and colleagues developed a semi-analytical solution for the eVDF at the center of a 1D finite-size plasma slab undergoing expansion [58, 59]. Leveraging the fact that the self-similar potential field evolves slowly in comparison to the electron transit time, the electron kinetics within the 1D finite-size plasma slab expansion can be described using the adiabatic invariant theory. As shown in [58, 59, 62], the energy increase experienced by those electrons with a sufficiently low initial energy is due to the adiabatic compression by the self-similar potential field, a phenomenon also observed by Khrabrov *et al* [60]. At the same time, those electrons with an initial energy in the intermediate range are reflected from the receding edge, leaving them with a reduced energy. This increases the population in the low-energy segment of the eVDF. As a result of the combined effect from the compression of the very low energy electrons and the reflection of the intermediate energy electrons, the mean energy of the low-energy segment of the eVDF increases. Those electrons with a sufficiently high initial energy can escape the potential barrier. This reduces both the population and the mean energy in the high-energy segment of the eVDF. Consequently, the eVDF at the center of the plasma slab start to deviate from the initial Maxwellian VDF, showing an increase in the low energy population and a depletion in the high-energy population. These features are similar to the non-Maxwellian features shown in figure 12.

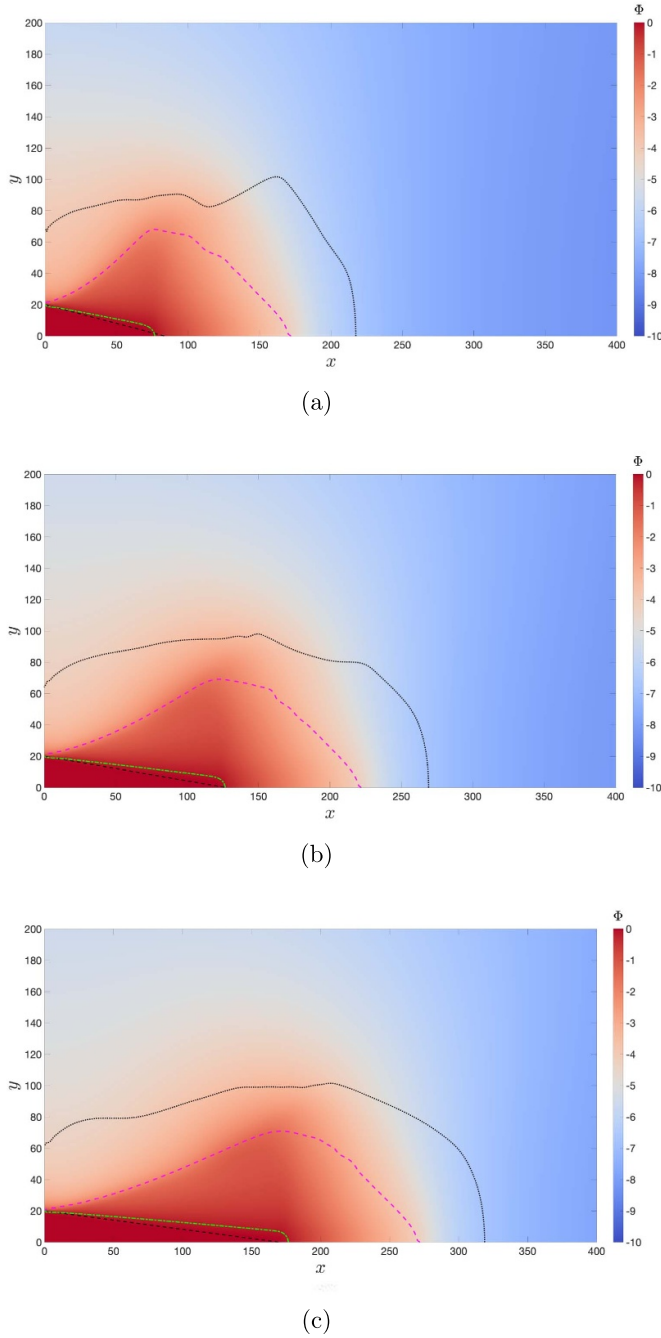


Figure 8. Electric potential contours at $t = 1000$ for cases with different initial Mach number. Black dashed line: Mach line. Green dash-dotted line: iso-contour line of the electron density value at the on-axis rarefaction wave front. Magenta dashed lines: iso-contour lines of the ion density value at the on-axis quasi-neutral fronts and stands for the ion beam front. Black dotted line: iso-contour line of $n_e \sim 0$. (a) $M_0 = 4.29$. (b) $M_0 = 6.43$. (c) $M_0 = 8.58$.

We next compare previous analytical solutions of eVDF with our numerical results of $\hat{f}_e(v_y)$ in 2D beam expansion. Khrabrov *et al* [60] presented a more rigorous analytical solution but the implicit form of the solution makes the comparison more complicated. On the other hand, Mora and Grismayer's derivation utilizes two first order approximations to obtain an

explicit form of the solution [58]. Here, we compare our results with Mora's explicit semi-analytical solution [58]:

$$f_e(v, t) = \left[1 + \left(\alpha - \beta \frac{v^2}{2} \right) \frac{v^2}{2} \frac{t}{\tau} \right] f_{e0} \quad (22)$$

where $\tau = L/C_{s0}$, L is the length of the initial plasma slab, and f_{e0} is the initial eVDF at $t = 0$. To apply Mora's solution, we correlate the eVDFs at various on-axis locations at $t\omega_{pe0} = 1000$ in our simulations with the eVDF at the center point of Mora's 1D finite-size plasma slab expansion model at different times. We introduce a mapping parameter $t^* = x/v_d$, where t^* denotes the evolution time in Mora's 1D finite-size plasma slab expansion model and x denotes the axial location in the 2D plasma beam expansion at $t\omega_{pe0} = 1000$. We replace the time t in Mora's solution with the mapping parameter t^* , set $\tau = y_{\text{beam}}/C_{s0}$, where y_{beam} is the y direction distance between the ion beam edge to the axis, and take f_{e0} to be the 1D eVDF at the injection plane, $\hat{f}_e(v_y)(x = 0)$. Thus, Mora's solution can be re-written as

$$\begin{aligned} \hat{f}_{\text{Mora}}(v_y) &= \int \hat{f}_e dv_x = \left[1 + \left(\alpha - \frac{\beta v_y^2}{2} \right) \cdot v_y^2 \cdot \frac{t^*}{\tau} \right] \cdot \int \hat{f}_{e0} dv_x, \\ \tau &= \frac{L}{C_{s0}}, \\ \int \hat{f}_{e0} dv_x &= \left(\frac{m_e}{2\pi T_{e,y}(0,0)} \right)^{1/2} \exp \left(-\frac{m(v_y - u_{e,y}(0,0))^2}{2T_{e,y}(0,0)} \right) \end{aligned} \quad (23)$$

where α and β are the control parameters that influence the shape and characteristics of the distribution function. We find $\alpha = \beta = 1$ provides the best fit of our simulation result to equation (23).

In figures 12(a)–(d), we compare $\hat{f}_e(v_y)$ with the fitted $\hat{f}_{\text{Mora}}$. We find that $\hat{f}_e(v_y)$ aligns well with the fitted $\hat{f}_{\text{Mora}}$ in figures 12(a)–(c) in the low to medium energy range, as predicted by equation (23). However, $\hat{f}_e(v_y)$ deviates from $\hat{f}_{\text{Mora}}$ at the high energy range, suggesting that the high energy electrons do not follow the same behavior as predicted by equation (23). This is because Mora's semi-analytical solution is derived by adiabatically calculating the variation of the electron energy for the electrons that can be trapped in the linear self-similar potential field formed at the edge of the plasma slab before the inward-marching rarefaction wave reaches the center. Thus, the semi-analytical solution equation (23) does not correctly describe the dynamics of those high energy electrons that escape the linear potential field [58]. In the 2D beam expansion considered here, some of the high-energy electrons will escape the potential field, resulting in a depleted high energy tail in the $\hat{f}_e(v_y)$, which is not described by equation (23). At the same time, some of the intermediate-energy electrons will lose energy due to reflection from the edge of the potential well in the transverse direction, while the sufficiently low-energy electrons will gain energy due to adiabatic compression by the potential field. Similar to that

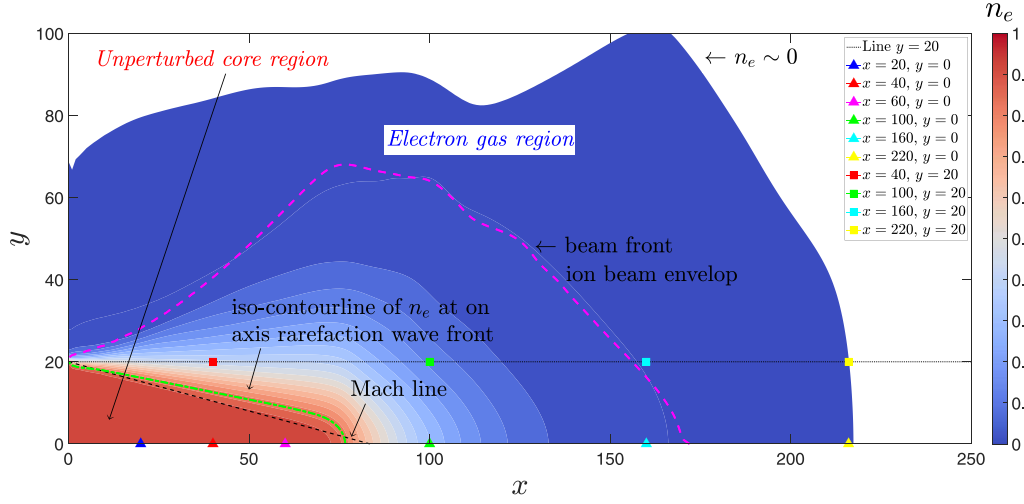


Figure 9. Different regions in the beam and the the selected locations for plotting eVDF, with n_e density contours, the Mach line, and the beam envelope. The iso-contour line of n_e for the density of the on-axis rarefaction wave front is shown by the green dash-dotted line.

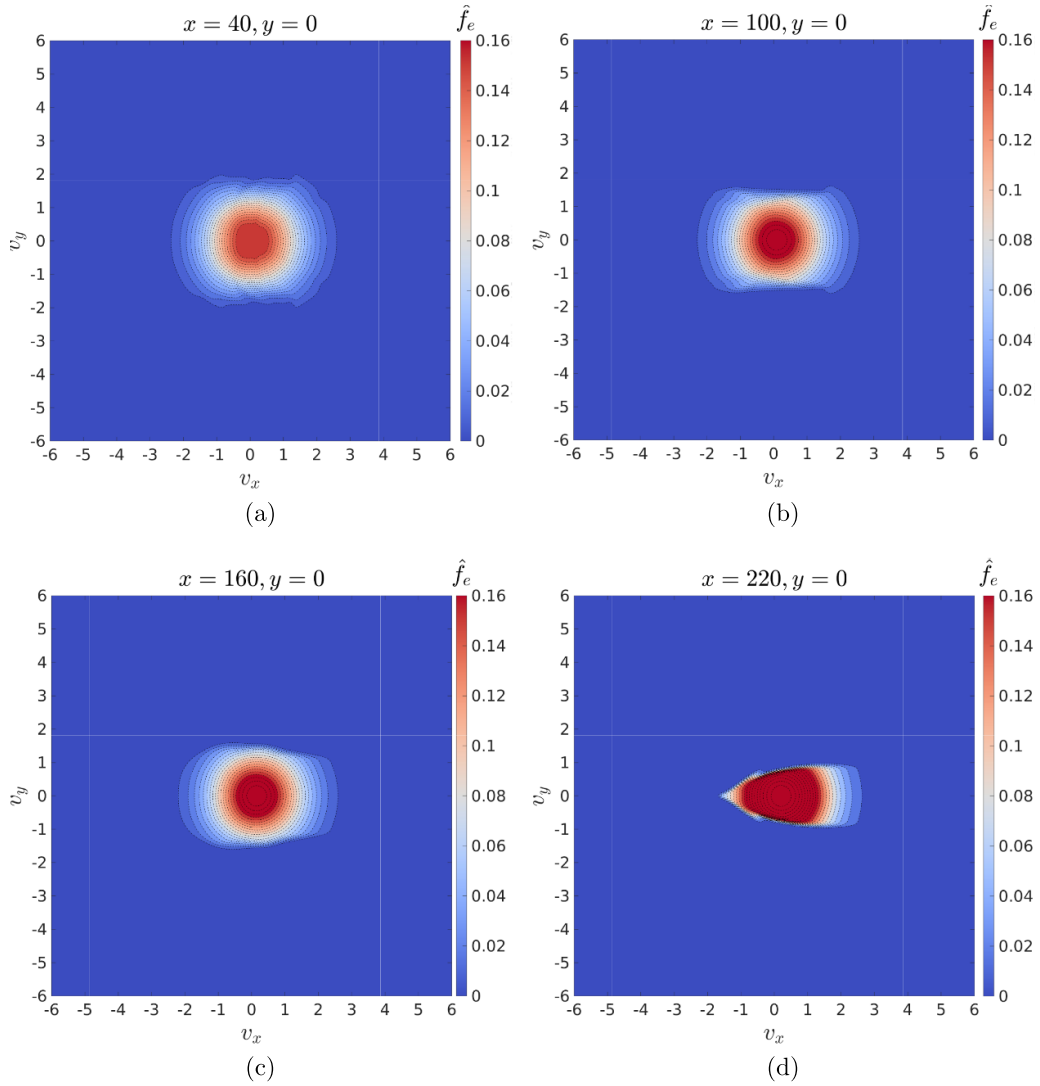


Figure 10. Density normalized electron velocity distribution function contours at $t = 1000$. (a) $(x, y) = (40, 0)$, a position selected from the region in the ‘unperturbed’ beam core region. (b) $(x, y) = (100, 0)$, a position inside the self-similar region. (c) $(x, y) = (160, 0)$, a position near ion beam front calculated by equation (20). (d) $(x, y) = (220, 0)$, a position in beyond ion beam front.

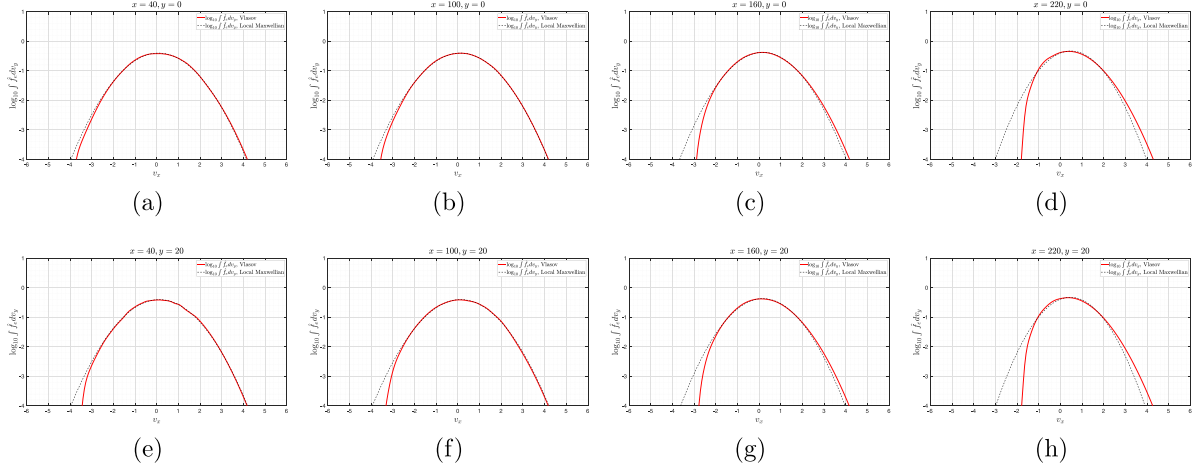


Figure 11. Density normalized v_x direction electron velocity distribution function plots at $t = 1000$. (a)–(d) On axis ($y = 0$) positions with $x = 40, x = 100, x = 160, x = 220$ respectively. (e)–(h) positions along $y = 20$ with $x = 40, x = 100, x = 160, x = 220$ respectively. The black dashed lines in all figures are the local reconstructed Maxwellian distribution.

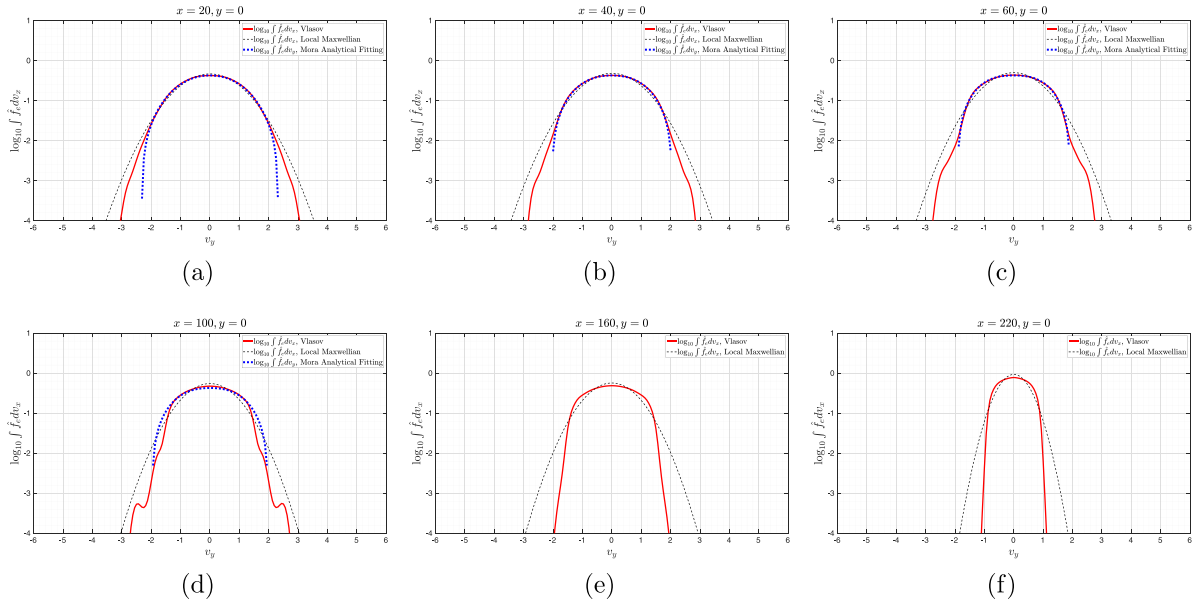


Figure 12. Density normalized v_y direction electron velocity distribution function plots at $t = 1000$. (a)–(f) On axis ($y = 0$) positions with $x = 20, x = 40, x = 60, x = 100, x = 160, x = 220$ respectively. The black dashed lines in all figures are the local reconstructed Maxwellian distribution. The blue dotted lines in (a)–(d) are the semi-analytical fitted distribution functions by equation (23). Note that only the positive part of the fitted distribution function is shown.

in the 1-D finite-size plasma slab expansion, the combination of these effects leads to an increased population in the low-energy range, resulting in a flattened top in the middle of $\hat{f}_e(v_y)$.

The ‘top-hat’ (i.e. the super Maxwellian distribution) $\hat{f}_e(v_y)$ is similar to the eVDF in 1D finite-size plasma expansion. According to Mora’s study, the two inward-marching rarefaction waves in the transverse direction meet at the position $x = v_d \cdot t^{\star'} = v_d \cdot 0.88\tau \approx 75$ [58]. Beyond this point, equation (23) can no longer accurately describes the evolution of the eVDF. This observation is evident in figure 12(d), where the fitted $\hat{f}_{\text{Mora}}$ deviates significantly from the actual distribution function. At that location, all electrons are losing energy,

and the eVDF further evolves towards a super Maxwellian distribution.

3.4. Electron heat flux

In almost all EP plume models, the electrons are simplified to be a massless ideal gas undergoing an assumed thermodynamic process. Under this assumption, a simplified model for electrons can be provided either by the Boltzmann relation for the isothermal process or the more general polytropic law. Note that the isothermal process ($\gamma_e = 1$) corresponds to a heat flux of $Q_e = \infty$ while the adiabatic process ($\gamma_e = 5/3$) corresponds to a heat flux of $Q_e = 0$. Previous studies [11,

19, 63] have attempted to establish a pseudo polytropic law using the electron temperature and density along the beam axis. However, as the electron thermodynamic process was not identified, the value of γ_e was arbitrary.

In order to understand the thermodynamic process for the electrons, one first needs to resolve the electron heat flux. Figure 13(a) shows the contour of the x direction electron heat flux Q_{ex} . Additionally, the green dash-dotted line shows the electron density contour line at a value equaling to that at the on-axis location of the rarefaction wave front, and the magenta dashed line shows the ion density contour line at a value equaling to that at the on-axis location of the quasi-neutral beam front given by equation (20). We find that the Q_{ex} contours exhibit three distinct regions. The first region is a triangular shaped region located next to the beam exit where Q_{ex} is near constant. This region maintains its a quasi-steady shape over time. Adjacent to the first region is a fan-shaped region which expands along the beam direction. Within this fan-shaped region, Q_{ex} increases with time and eventually becomes comparable to that in the first region. The outer boundary of the fan-shaped region matches approximately the green dashed-dotted iso-contour line. Figure 13(a) shows that $Q_{ex} \simeq 0$ beyond the ion beam boundary ($n_i \simeq n_e \simeq n_{\text{front}}$).

Figure 13(b) shows the contour of the y direction electron heat flux Q_{ey} . Similar to figure 13(a), the magenta dashed line approximately separates Q_{ey} into a non-zero region and a near zero region. The Q_{ey} component demonstrates a relatively high amplitude in the vicinity of the injection surface, comparable to the magnitudes of Q_{ex} observed in the core and fan regions in figure 13(a). Consequently, this region can be identified as the core region for Q_{ey} . In the region outside the core region (but still within the plume), the magnitude of Q_{ey} is approximately one order of magnitude smaller compared to that observed in the core region. From figures 13(a) and (b), it can be observed that Q_{ey} plays a significant role only in the core region near the injection plane. In the other regions within the plume, Q_{ex} is the dominant component of the total electron heat flux.

For the 2D setup investigated in this study, we further express Q_{ex} in the sum of two electron heat flux tensors elements

$$\begin{aligned} Q_{ex} &= \iint \frac{1}{2} m_e (v_x - \langle v_x \rangle)^2 (v_{ex} - \langle v_{ex} \rangle) f_e dv_x dv_y \\ &= \iint \frac{1}{2} m_e \left[(v_{ex} - \langle v_{ex} \rangle)^2 + (v_{ey} - \langle v_{ey} \rangle)^2 \right] \\ &\quad \times (v_{ex} - \langle v_{ex} \rangle) f_e dv_x dv_y \\ &= \frac{1}{2} \iint m_e (v_{ex} - \langle v_{ex} \rangle)^3 f_e dv_x dv_y \\ &\quad + \frac{1}{2} \iint m_e (v_{ey} - \langle v_{ey} \rangle)^2 (v_{ex} - \langle v_{ex} \rangle) f_e dv_x dv_y \\ &= 0.5 \cdot \bar{\bar{Q}}_{e,xxx} + 0.5 \cdot \bar{\bar{Q}}_{e,yyx}. \end{aligned} \quad (24)$$

Figure 14 shows Q_{ex} as well as $\bar{\bar{Q}}_{e,xxx}$ and $\bar{\bar{Q}}_{e,yyx}$. It is observed that $\bar{\bar{Q}}_{e,xxx}$ exhibits a feature similar to Q_{ex} at a short distance downstream from the injection plane. The magnitude of $\bar{\bar{Q}}_{e,xxx}$

is about 2–6 times larger than the magnitude of $\bar{\bar{Q}}_{e,yyx}$ in this region. The $\bar{\bar{Q}}_{e,yyx}$ only has a large contribution to the Q_{ex} in the region adjacent to the injection plane and remains negligible in the region downstream. Hence, it can be concluded that $\bar{\bar{Q}}_{e,xxx}$ has the major contribution to the total electron heat flux, and thus, dominates electron energy transfer in the plume expansion process. The on-axis profiles of the heat flux vector and tensor also show that the heat transfer process considered here does not follow the Fourier type heat conduction law, as in classical thermodynamics, since the x direction electron temperature T_{ex} remains quasi-constant inside the plume and the near-zero $\partial T_{ex} / \partial x$ can not explain the large heat flux shown in figure 14.

3.5. Discussions

Because of the dominance of the $\bar{\bar{Q}}_{e,xxx}$ component in the electron energy transfer process, the subsequent discussions pertaining to the connection between electron microscopic kinetic features and macroscopic collisionless electron heat flux will focus on the $\bar{\bar{Q}}_{e,xxx}$.

As shown in figure 11, the eVDFs normalized by density in the v_x direction exhibit a similar degree of depletion in the high-energy tail of the negative branch at various positions within both the upstream ‘unperturbed’ region and the quasi-neutral beam region. This indicates that the eVDFs in these regions share similar skewed shapes. In statistics, the skewness $\mathbb{S}$ is a measure used to quantify the asymmetry of a distribution. It provides information about the relative sizes of the two tails in the distribution function [64]

$$\mathbb{S}[V] = \mathbb{E} \left[\left(\frac{V - \mu}{\sigma} \right)^3 \right] = \int \left(\frac{v - \mu}{\sigma} \right)^3 g dv \quad (25)$$

where V is the random variable, $\mathbb{E}$ is the expectation operator. The expectation of V , μ is defined as $\mu = \mathbb{E}[V] = \int v g dv$. The standard variation σ is defined as $\sigma = \sqrt{\mathbb{E}[(V - \mu)^2]} = \sqrt{\int (v - \mu)^2 g dv}$. The probability function $g(v)$ satisfies the property $\int g dv = 1$. If one considers that $g = \int \hat{f}_e dv_y$ and $V = v_x$, the statistical moments μ and σ can be interpreted as x electron direction bulk velocity $u_{e,x}$ and x direction electron thermal velocity $v_{te,x}$ respectively. The skewness of the v_x direction marginal eVDF can then be written as

$$\begin{aligned} \mathbb{S}[v_x] &= \int \left(\frac{v_x - u_{e,x}}{v_{te,x}} \right)^3 \left(\int \hat{f}_e dv_y \right) dv_x \\ &= \frac{1}{n_e m_e v_{te,x}^3} \iint m_e (v_x - u_{e,x})^3 f_e dv_x dv_y \\ &= \frac{\bar{\bar{Q}}_{e,xxx}}{n_e m_e v_{te,x}^3}. \end{aligned} \quad (26)$$

Equation (26) expresses the skewness of the marginal eVDF in the v_x direction as the ratio between the heat flux tensor component $\bar{\bar{Q}}_{e,xxx}$ and the local variable $n_e m_e v_{te,x}^3$. This equation establishes a relationship between the collisionless heat flux

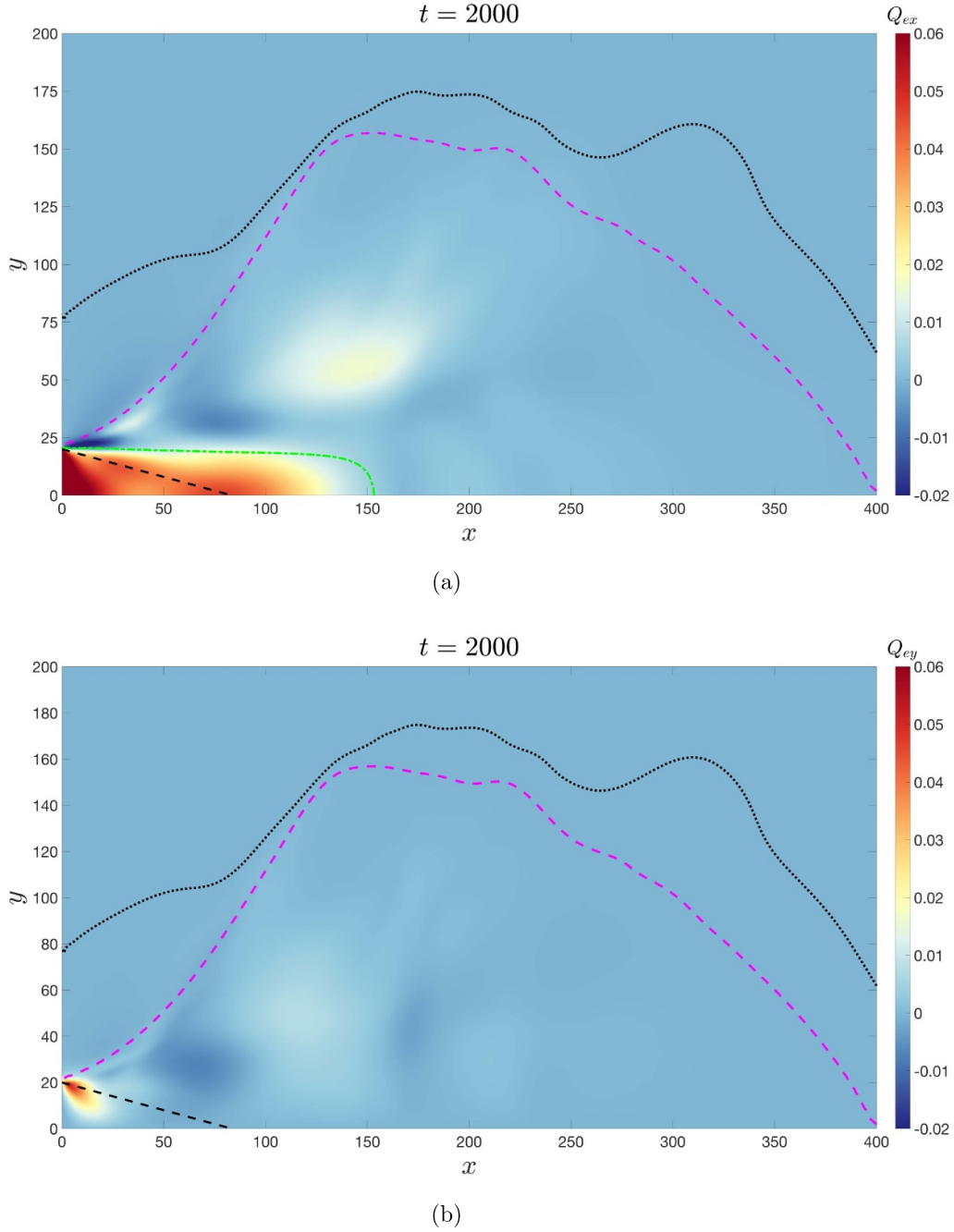


Figure 13. Contour of electron heat flux at $t = 2000$. Black dashed line: Mach line. Green dash-dotted line: iso-contour line of the electron density value at the on-axis rarefaction wave front. Magenta dashed lines: iso-contour lines of the ion density value at the on-axis quasi-neutral fronts and stands for the ion beam front. Black dotted line: iso-contour line of $n_e \sim 0$. (a) Q_{ex} . (b) Q_{ey} .

tensor $\bar{\bar{Q}}_{e,xxx}$ and the relative asymmetry in the tail of eVDF. The right-hand side of equation (26) corresponds to the coefficient α in the flux-limited closure model for collisionless electron heat flux, which has been utilized in studies related to the solar wind [15] and fusion scrape-off layer research [65]. This suggests that studying the relative asymmetry of eVDF can aid in selecting the appropriate coefficient for establishing a potential collisionless heat flux closure model.

Figure 15(a) plots $\bar{\bar{Q}}_{e,xxx}/(nm_e v_{te,x}^3)$ (i.e. the flux-limited model coefficient) as a function of distance along the beam axis at different times. It is observed that, after a short distance

from the injection plane, $\bar{\bar{Q}}_{e,xxx}/(nm_e v_{te,x}^3)$ exhibits a time-dependent feature consisting of a hump and a nearly monotonically increasing tail. Interestingly, these structures are self-similar feature at different times. This suggests that the propagation of $\bar{\bar{Q}}_{e,xxx}/(nm_e v_{te,x}^3)$ along the beam axis exhibits certain patterns that remain consistent over time. Based on these distinct self-similar features, $\bar{\bar{Q}}_{e,xxx}/(nm_e v_{te,x}^3)$ can be split into two distinct regions, region A and B, as shown in figures 15(b) and (c). The boundary between region A and region B is represented by the red dashed line, which is defined as $x = t(v_d - C_s)$, corresponding to the location of the ion

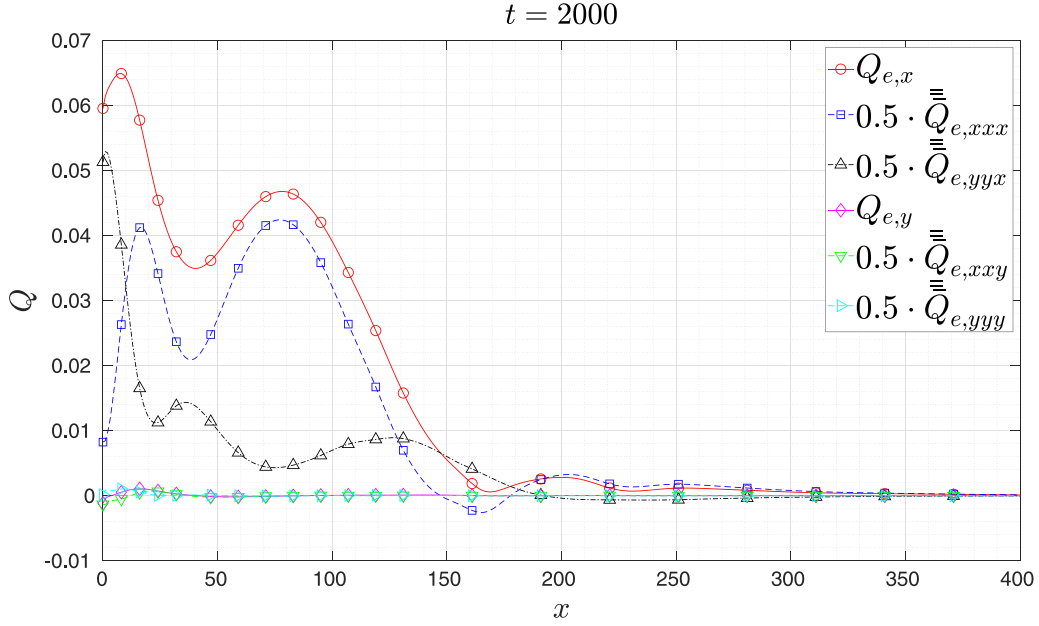


Figure 14. $Q_{e,x}$, $\bar{Q}_{e,xxx}$, $\bar{Q}_{e,yyx}$, $Q_{e,y}$, $\bar{Q}_{e,xyy}$, $\bar{Q}_{e,yyy}$ profiles along axis at $t = 2000$.

rarefaction wave front at the axis. The hump structure is confined within region A. Region B is situated downstream of region A, and partially overlaps with the quasi-neutral beam region. $\bar{Q}_{e,xxx}/(nm_e v_{te,x}^3)$ increases nearly monotonically along the beam in this region.

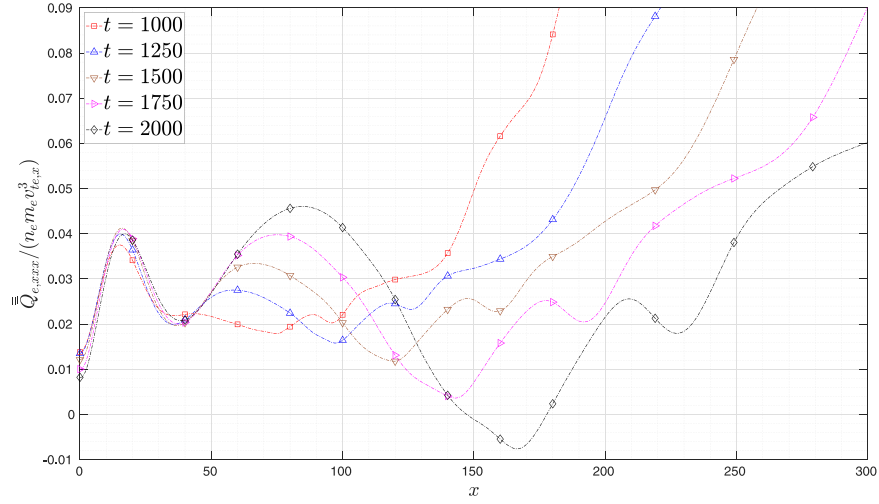
The increasing trend of $\bar{Q}_{e,xxx}/(nm_e v_{te,x}^3)$ in region B can be attributed to the following: in the self-similar region, a quasi-constant electric field arises due to the quasi-neutrality of the net charge, which results in a negligible space charge density along the beam direction [56, 57, 61]. As the electrons are pulled upstream by the electric field, a significant portion undergoes reflection. As the beam progresses downstream, only those electrons with sufficiently large positive velocities are able to pass through. Consequently, the skewness in the eVDF in the x direction increases due to the depletion of the negative velocity tail. This phenomena is reflected in the increasing flux limited coefficient observed in region B. The validity of this process is demonstrated in figures 10(b)–(d) and 11(b)–(d).

The skewness in eVDF is primarily influenced by the electron kinetics along the beam direction. The effect of the electron dynamics and the electric field in the transverse direction on the electron dynamics in the beam direction can be considered as providing an ‘effective potential’. Utilizing an analogy to the 1D electron par-axial kinetic model for magnetized plume expansion [66, 67], the electrons’ transverse action integral $J = \oint m_e v_y dy$ can be considered as an adiabatic invariant because the change of J only has a second order dependence on the beam divergence angle [67]. The impact of transverse electron motion on the expansion of an unmagnetized beam is similar to the magnetic mirror phenomena observed in magnetized plume expansions [68]. This similarity arises because the conservation of the adiabatic invariant J results

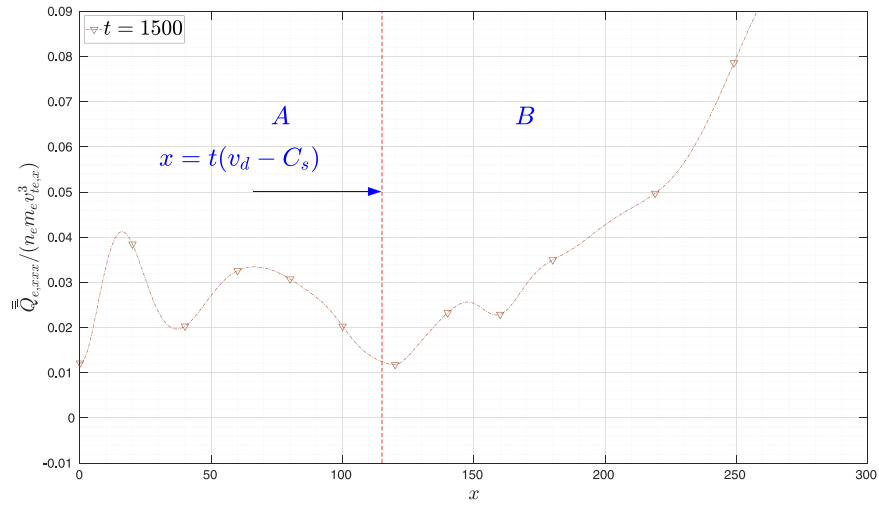
in the acceleration of electrons downstream, while the electric potential field acts to decelerate them.

Previously, Wang *et al* [8] studied the dynamics of the electrons in an unmagnetized, collisionless mesothermal plasma beam using full particle PIC, and showed that electron–ion coupling in a collisionless beam is achieved through interactions between electrons and a potential well established between the emission surface and the propagation of the ion beam front along the beam direction. The dynamics of individual electrons can also be obtained from Vlasov simulation. Figure 16 shows a representative electron trajectory. This trajectory is obtained by post-processing the potential data from the grid-based Vlasov simulation and tracing a test particle in the evolving electric field from $t=979$ to $t=2000$ following the method in [38]. Figure 16(a) shows the electron trajectory in the physical space (x, y) . The trajectory clearly shows the trapping of the electron in both the beam and the transverse direction. The beam region containing this test electron can be characterized as slowly diverging, and thus the paraxial approximation for this test electron is valid. Focusing on the dynamics along the beam direction, figure 16(b) shows the electron trajectory in the phase space (x, v_x) . Consistent with the findings presented in [8], our results show that a competition between electron acceleration resulting from the conservation of the electron’s transverse motion integral and electron deceleration due to the electric potential leads to the trapping of certain electrons within the beam.

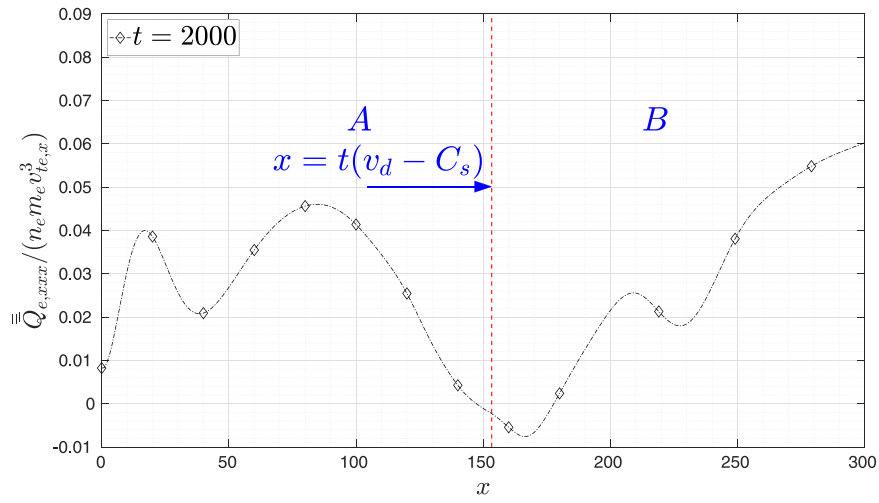
The asymmetric feature of the eVDF has long been recognized as the cause of collisionless heat flux in the solar wind corona. However, the mechanisms underlying the formation of such asymmetric features remain unclear. Perkins [69] and Hollweg [15] developed flux-limited models to describe the



(a)



(b)



(c)

Figure 15. (a) Flux limited model coefficient along axis at various time. (b) Region division at $t = 1500$. (c) Region division at $t = 2000$. In (b) and (c), the red dashed line is $x = t(v_d - C_s)$.

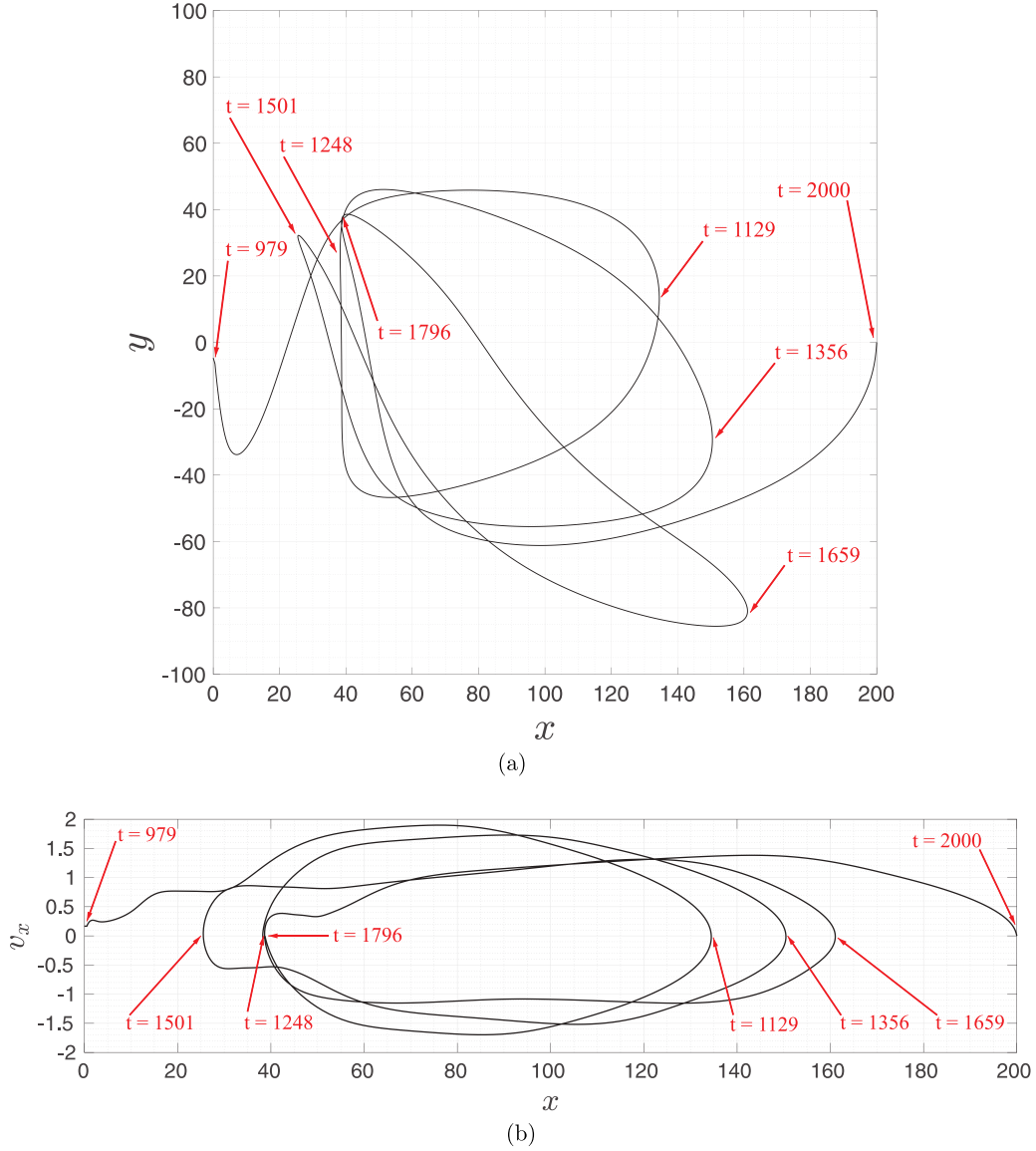


Figure 16. Representative electron trajectory. The trajectory is plotted for an electron with an initial condition of $(x, y) = (200, 0)$, $(v_x, v_y) = (0, 0.9)$ from time $t = 979$ to time $t = 2000$. The electron positions at several different time steps are marked on the trajectory to guide the eye. (a) Representative electron trajectory in physical space (x, y) . (b) Representative electron trajectory in phase space (x, v_x) .

collisionless heat flux in the solar wind corona and made contributions to elucidating the mechanisms behind this asymmetry. Their work suggests that this asymmetry is determined by the presence of the trapped electrons resulting from the competing processes of magnetic mirror effects and the monotonically decreasing electric potential. In the context of the unmagnetized plume expansion, the conservation of electron transverse direction motion plays a similar role to the magnetic mirror effects. Therefore, the trapping of electrons in region A can be considered a significant factor in determining the asymmetric part of the on-axis eVDF. The trapping loop of the electron trajectory shown in figure 16(b) exhibits a growing size over time while maintaining a similar loop shape. Hence, the qualitative explanation for the self-similar hump structures observed in region A can be attributed to the trapping of electrons in this region.

4. Summary and conclusions

The purpose of this paper is to present, to our knowledge, the first 2D grid-based Vlasov simulation model for an unmagnetized EP beam. Comparing to full particle PIC simulation, the Vlasov simulation approach successfully eliminates the interference from the particle noise inherent in the PIC method, and thus enables us to obtain accurate solutions on higher order velocity moments such as electron temperature and heat flux.

Vlasov simulations are carried out to study eVDF and electron thermodynamic properties in EP beam. The results show that the eVDF in v_x exhibits a near-Maxwellian shape with a depleted tail in the negative velocity branch at different positions within the beam. On the other hand, the eVDF in v_y evolves towards a top-hat shape. The evolution of eVDF in v_y follows Mora's semi-analytical solution for 1D finite-size

thin-foil plasma adiabatic expansion adiabatic solution. This suggests a near-adiabatic cooling process for the electrons in the transverse direction of the beam. The results further show that the magnitude of the electron heat flux in the beam is dominated by that from the x direction heat flux tensor $\bar{\bar{Q}}_{e,xxx}$.

We find that $\bar{\bar{Q}}_{e,xxx}$ and electron microscopic kinetics are connected through the skewness of the eVDF. Specifically, the equation for skewness of eVDF in v_x has the same form as the coefficient in the flux limited model of the electron collisionless heat flux. $\bar{\bar{Q}}_{e,xxx}$ shows a self-similar profile consisting of a hump and a tail. This profile can be further divided into two regions divided by the front of the backward propagating rarefaction wave in the beam direction. The behavior of $\bar{\bar{Q}}_{e,xxx}$ in the region upstream of the wave front is due to the trapped electrons, while that in the region downstream of the wave front is due to the presence of an acceleration of electrons by the strong electric field within the self-similar region. The connection between the eVDF and electron heat flux suggests a possible pathway to develop a thermodynamic closure relation that can account for microscopic kinetics for electrons in an unmagnetized beam expansion. The development of such a thermodynamic closure relation will be addressed in a future study.

Data availability statement

The data cannot be made publicly available upon publication because the cost of preparing, depositing and hosting the data would be prohibitive within the terms of this research project. The data that support the findings of this study are available upon reasonable request from the authors.

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